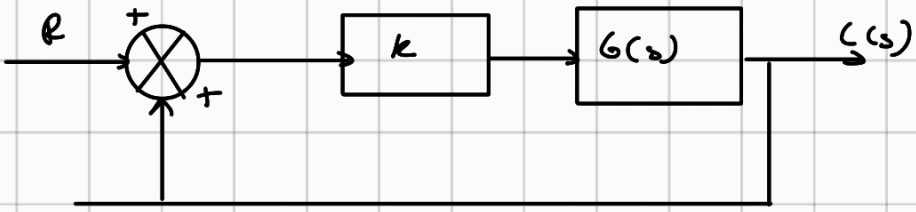
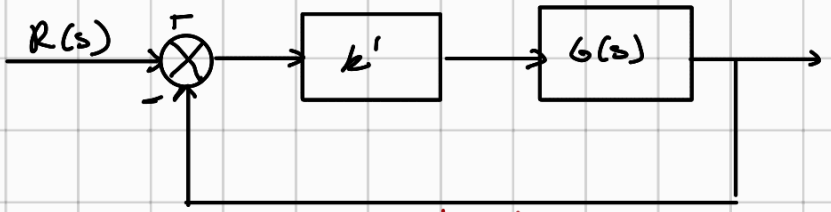
Positive Feed back architecture



$k > 0$



$k' < 0$

negative feedback

$$\frac{C(s)}{R(s)} = \frac{kG(s)}{1 - kG(s)H(s)}$$

$$1 - kG(s)H(s) = 0$$

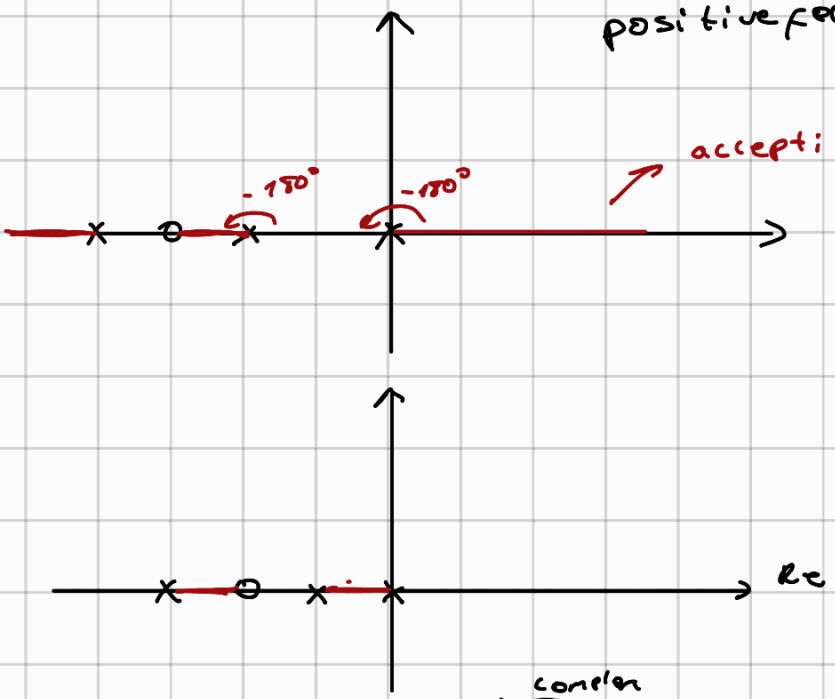
$$G(s)H(s) = \underbrace{\left(\frac{1}{k}\right)}_{\text{positive}} + 0j$$

$$\angle G(s)H(s) = \tan^{-1}\left(\frac{0}{\varepsilon^+}\right)$$

$$= \pm 360i \text{ or } \pm 180(2i)$$

$$\boxed{|G(s)H(s)| = \frac{1}{k}}$$

positive feedback



② Departure angles from (complex) poles

arrival " to zero (complex)

departure angle $= 360$ (180 degree) $+ \sum$ phase of cc pole w.r.t other pds $- \sum$ phase of cc pole w.r.t other zeros

③ Asymptote angles

$$\frac{\pm 360i}{n-m}$$

Example

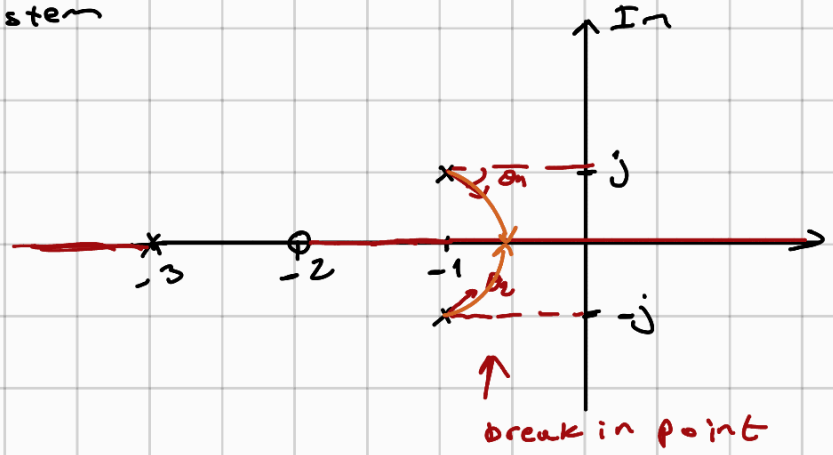
$$G(s) = \frac{k(s+2)}{(s+3)(s^2+2s+2)}$$

Positive feedback system

$z_1 = -2$

$p_1 = -3$

$p_{2,3} = -1 \pm j$



branches: 3

$n-m = 3-1 = 2$

angle of assymptotes : $\frac{\pm 360i}{n-m} = \pm 180i$

for $s = -1+j$

$\theta_1 = 0 - 27 - 90^\circ + 45^\circ = -72^\circ$

for $s = -1-j$

$\theta_2 = 0^\circ + 27^\circ + 90^\circ - 45^\circ = +72^\circ$

To find charackteristic equation

$$1 + \frac{k(s+2)}{(s+3)(s^2+2s+2)} = 0$$

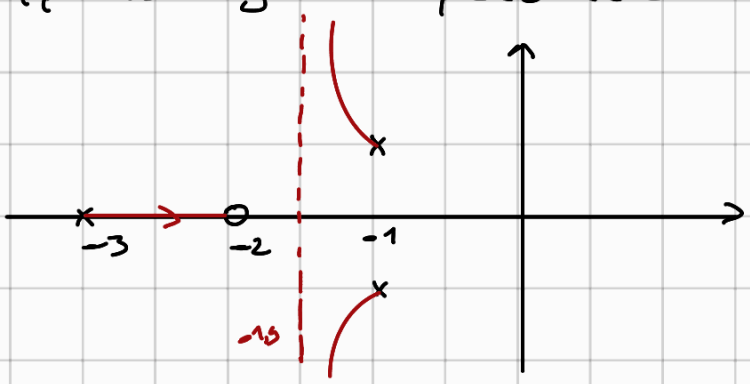
$$(s+3)(s^2+2s+2) + k(s+2) = 0$$

$$k = -\frac{(s+3)(s^2+2s+2)}{(s+2)}$$

$$\frac{dk}{ds} = 0 \Rightarrow 2s^3 + 11s^2 + 20s + 10 = 0 \Rightarrow 2(s+0,8)(s+2,35+j0,77)(s+2,35-j0,77) = 0$$

⇒ s = -0,8 breakin

if its negative feedback

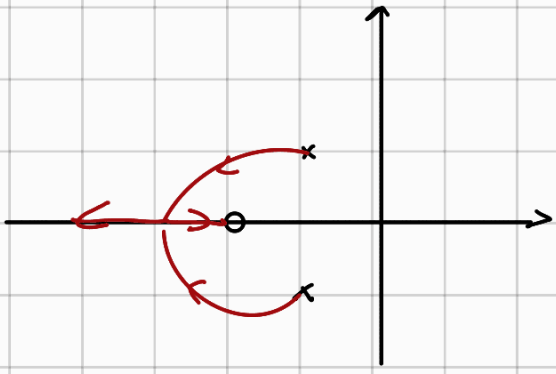


centroid = $\frac{\Sigma \text{real part poles} - \Sigma \text{real part of zeroes}}{n - m}$ = -1,5

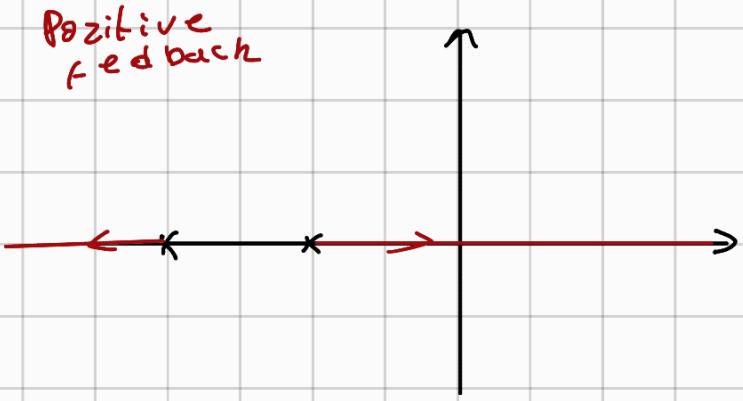
Example

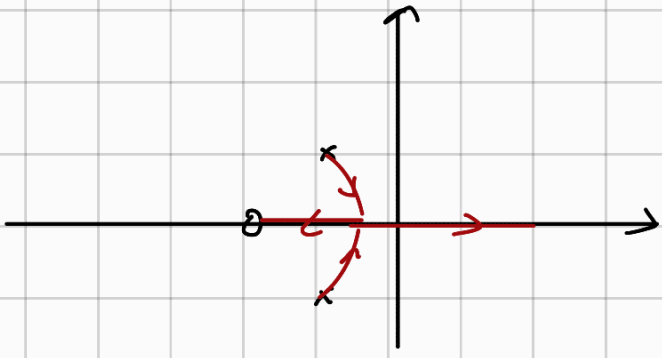
Negative feedback



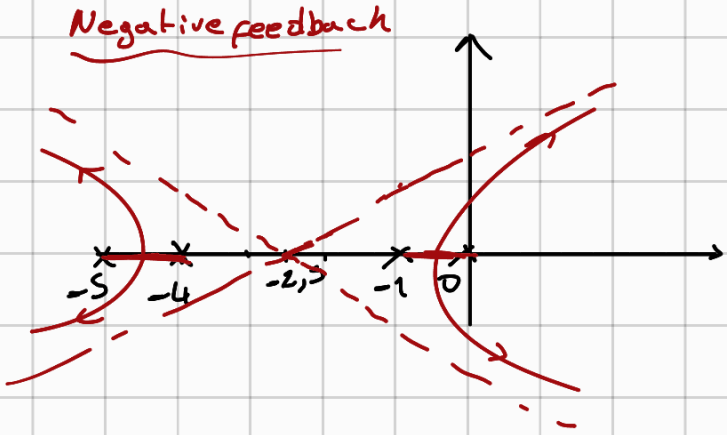


Pozitive feedback





Negative feedback



$\pm \frac{180(2i+1)}{n-m}$

$\pm 45(2i+1)$

$\pm 45 \quad \pm 135$

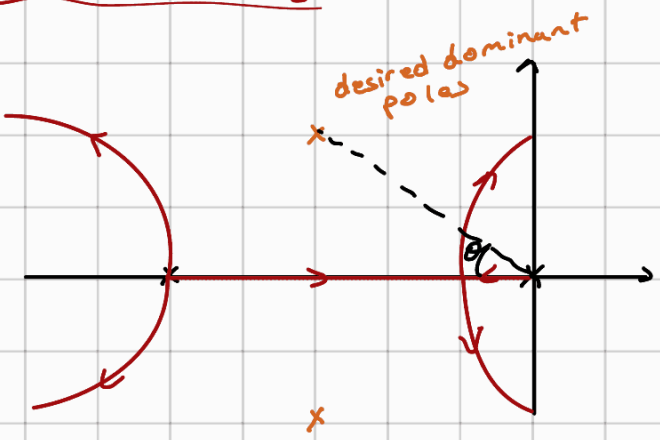
centroid = -2,5

pozitif f.



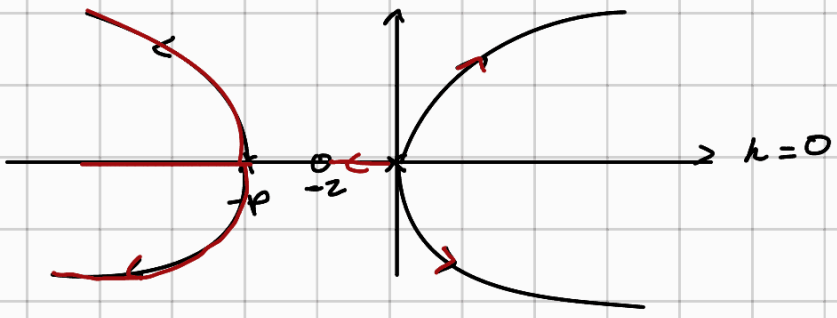
$\pm \frac{360i}{4} = \pm 90i$

Compensator design



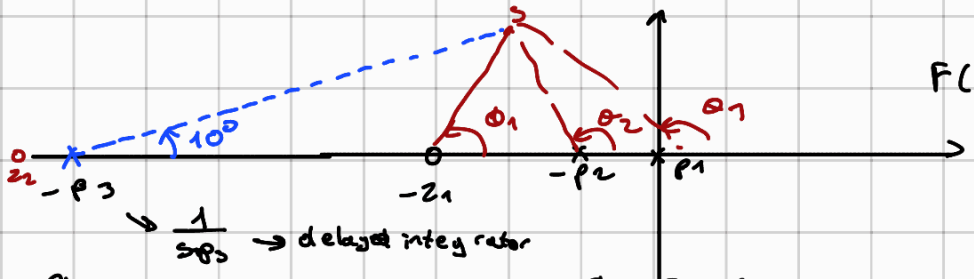
specific "k" which renders the system unstable

4 seperate roots of ch.eq.



$1 + \frac{k(s+2)}{s^3(s+p)^3}$

$\theta = \cos^{-1}\zeta$ damping ratio



$F(s) = \frac{s+a}{s+b} = \frac{Re^{i\phi}}{Me^{i\theta}} = \left(\frac{R}{M}\right)e^{i(\phi-\theta)}$

$\rightarrow \frac{1}{sp_3}$ → delayed integrator

Phase of 3 points $\beta = \theta_1 - \theta_2 - \theta_3$

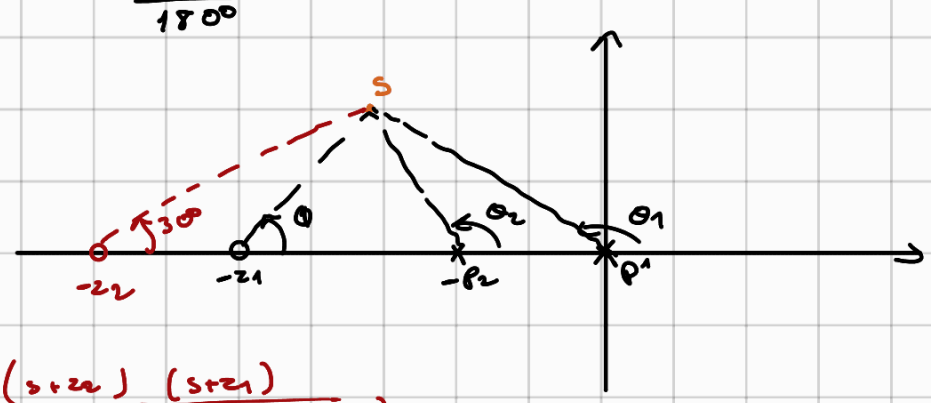
If $\beta = 190°$ → to make it negative $\frac{-10}{180}$ → p_3 angle have to be bigger

Negative

the original OLTF $\frac{s+z_1}{(s+p_1)(s+p_2)}$

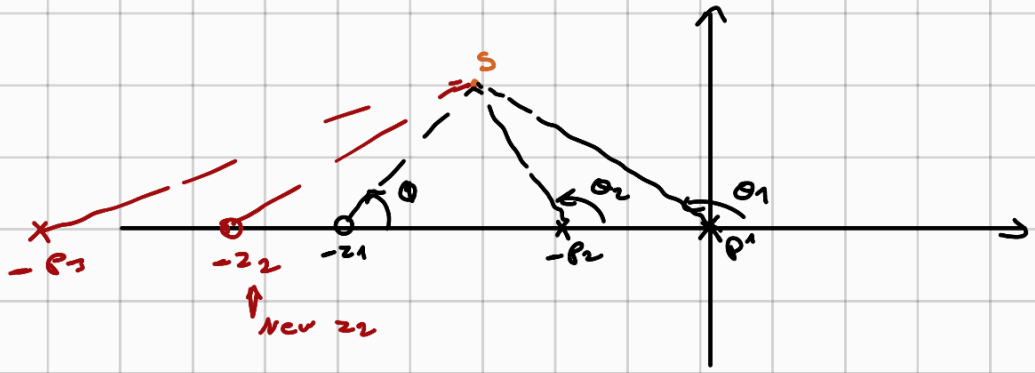
p_3 $\frac{s+z_2}{s+p_3} \simeq \frac{1+z_2}{\lim_{p_3 \to 0}}$

if $\beta = +\frac{150}{30}{}^{\circ}$ / 180°



$\frac{(s+z_2)}{(s+p_1)(s+p_2)}$

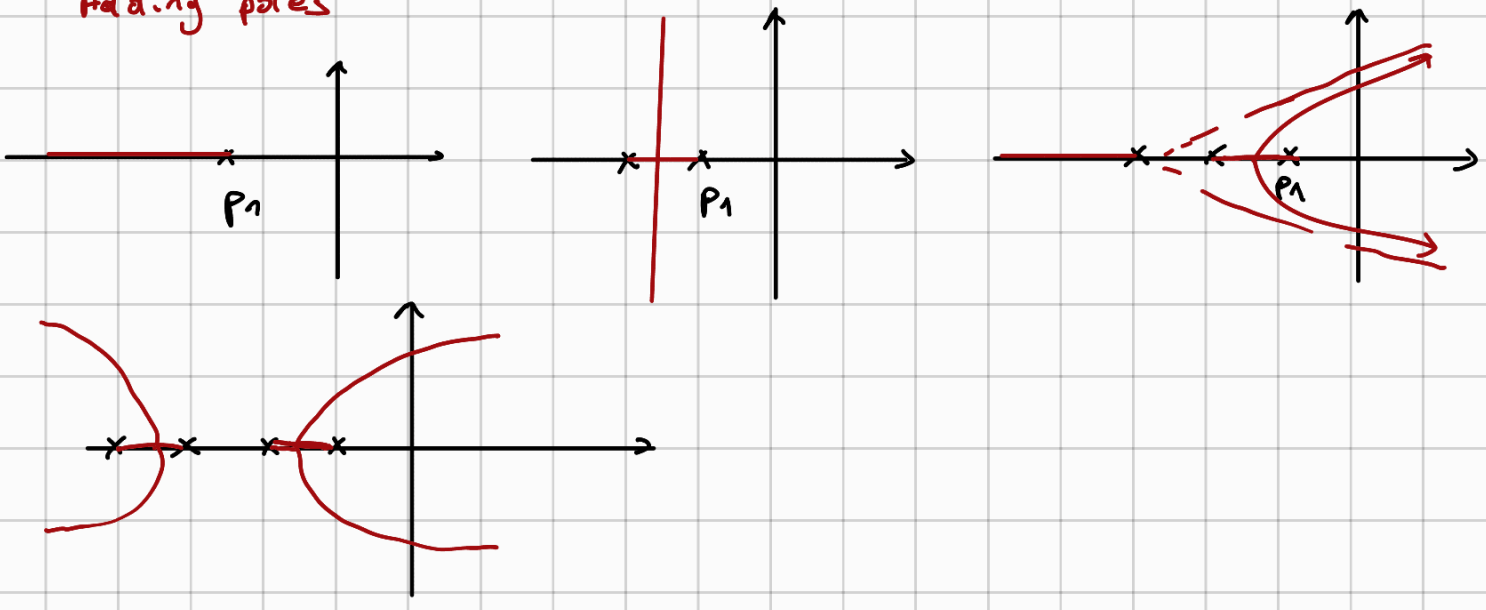
derivative operator
(ts unuseless itself so we have to add a pole



$\frac{s+z_2}{s+p_3} \lim_{z_2 \to 0} \frac{s}{s+p_3} = 1 \frac{-p_3}{s+p_3}$

?? WTF

Adding poles



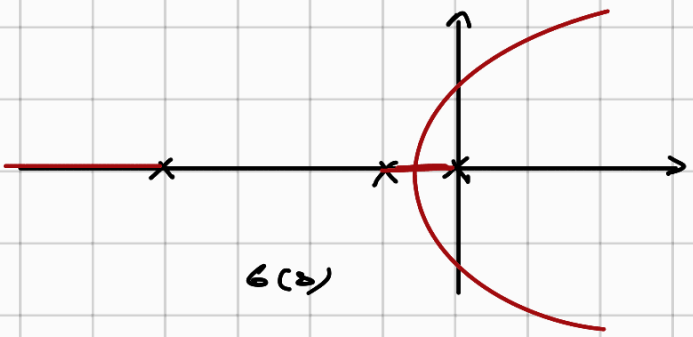
by adding poles the locus plot move to right systems get more unstable. (lag compensator)

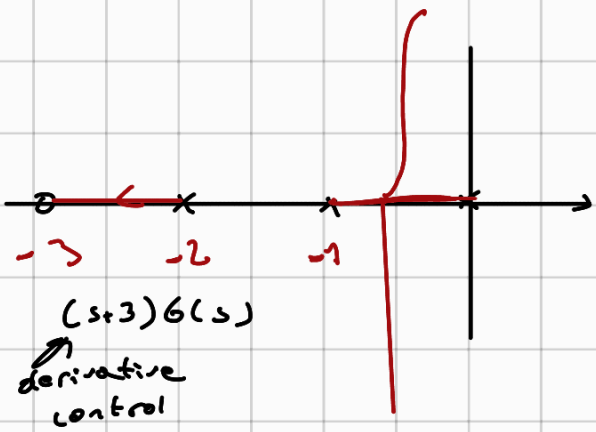
System get more unstable - slower!
($t_s \uparrow$)

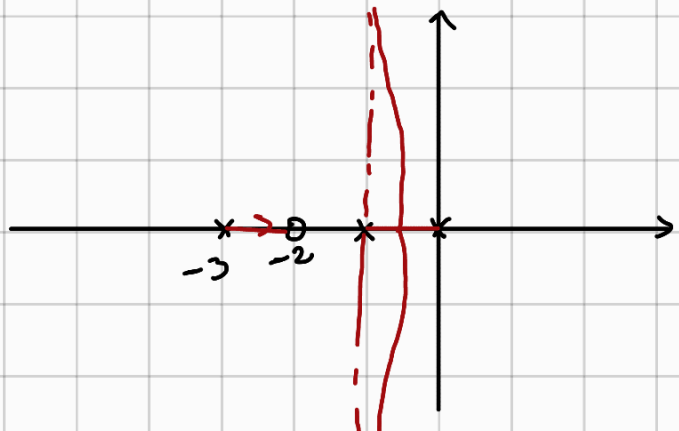
More accurate by adding integrator

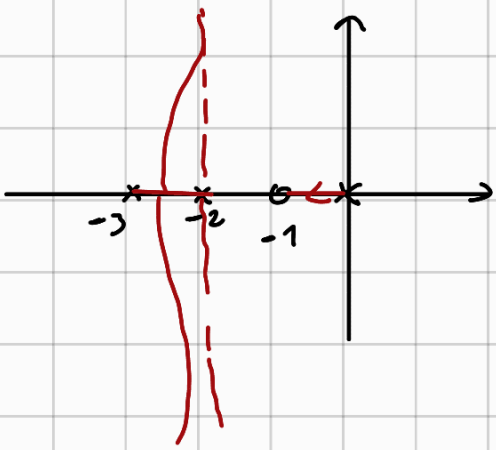
Lag compensation, The output angle is less than input

Zero placement









$\frac{-1-3-(-2)}{2} = \frac{-4+2}{2} = -1$

pulls the root locus plot to the left

↳ more stable

↳ faster response ($t_s \downarrow$)

more sensitive to noise

The output angle is more than the input angle

its leading output

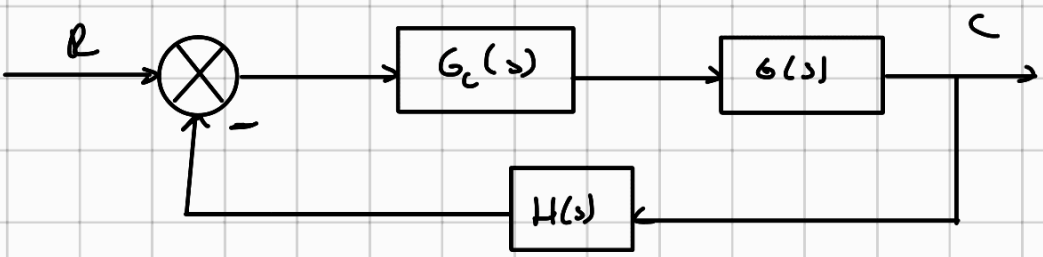
$r(t) = \sin \omega t \rightarrow y(t) = \sin(\omega t + \phi)$



lag compensation also

lead lagging compensator

Series / parallel compensation

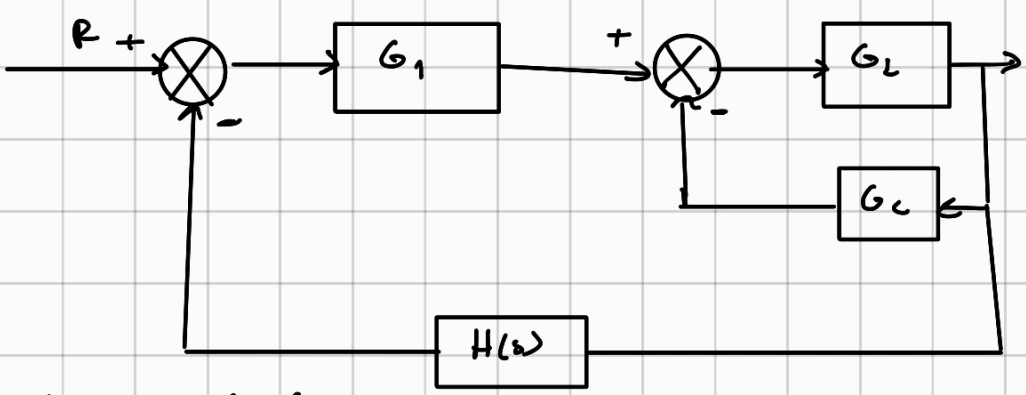


$$\frac{C(s)}{R(s)} = \frac{G_c G}{1 + G_c G H}$$

$1 + G_c G H = 0 \Rightarrow G_c G H = -1$

$G_c G H$ = gives the root locus

Parallel



$$\frac{C(s)}{R(s)} = \frac{G_1 G_2}{1 + G_2 G_c + G_1 G_2 H} \longrightarrow 1 + G_c \frac{G_2}{1 + G_1 G_2 H} = 0$$

$1 + G_2 G_c + G_1 G_2 H = 0 \Rightarrow 1 + G_c G_F = 0$

$\times \frac{1}{1 + G_1 G_2 H}$

$$= \frac{1 + G_1 G_2 H + G_2 G_c}{1 + G_1 G_2 H}$$

Lead compensator : zero is closer to jw axis

$$G_c(s) = k_c \frac{s+\frac{1}{T}}{s+\frac{1}{\alpha T}} \qquad 0<\alpha<1$$

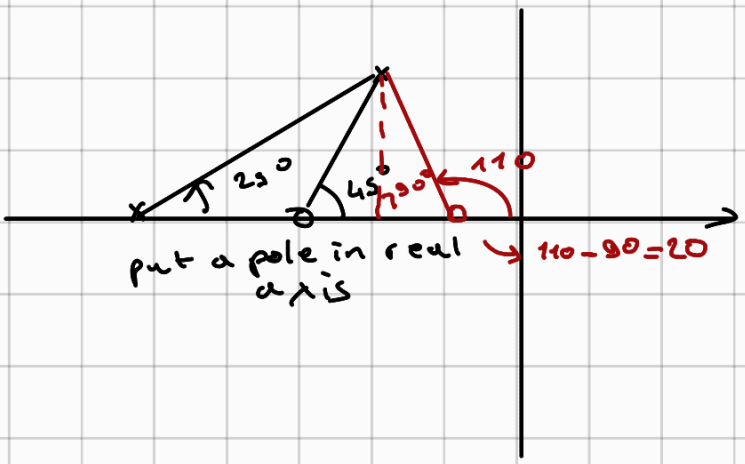
→ specs

$$k_c \frac{s+z}{s+p}$$

If $\alpha > 1 \Rightarrow$ lag compensator $\quad G_c(s) = k_c \dfrac{s+1/T}{s+\frac{1}{\beta T}} \qquad \beta > 1$

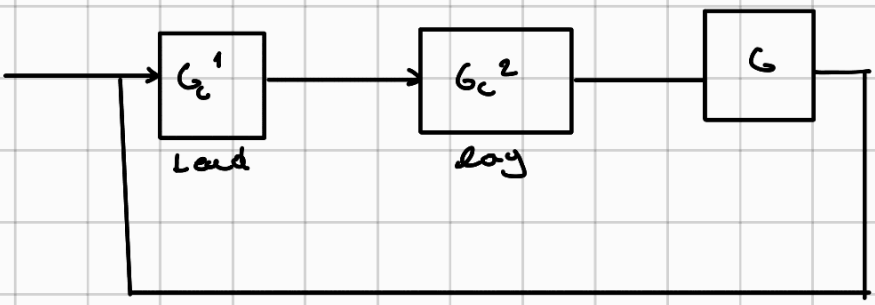
$\frac{1}{T} = 10$

$$\begin{cases} z = -10 \\ p = \frac{-10}{\alpha} = -20 \end{cases} \qquad \alpha = \frac{1}{2}$$



we have to make +20

lead lag compensator

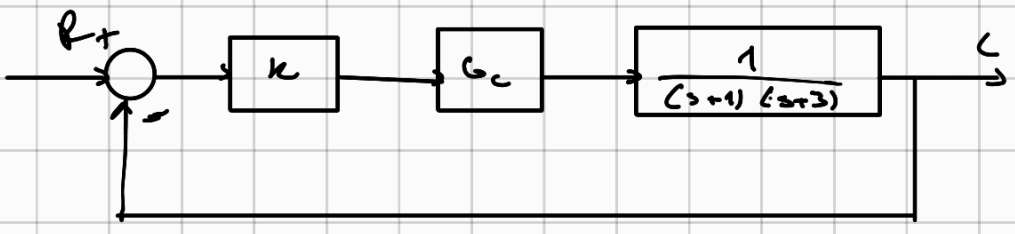


$$\left(k_c \frac{s+\frac{1}{T_1}}{s+\frac{1}{\alpha T_1}} \right) \left(\frac{s+\frac{1}{T_2}}{s+\frac{1}{\beta T_2}} \right)$$

S announce to be design

$0<\alpha<1 \qquad \beta > 1$

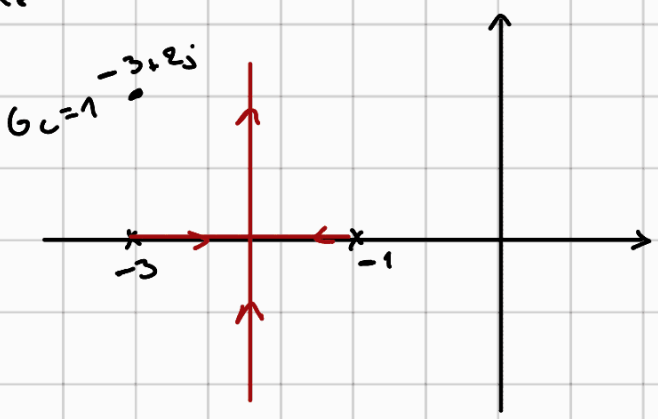
Ex :



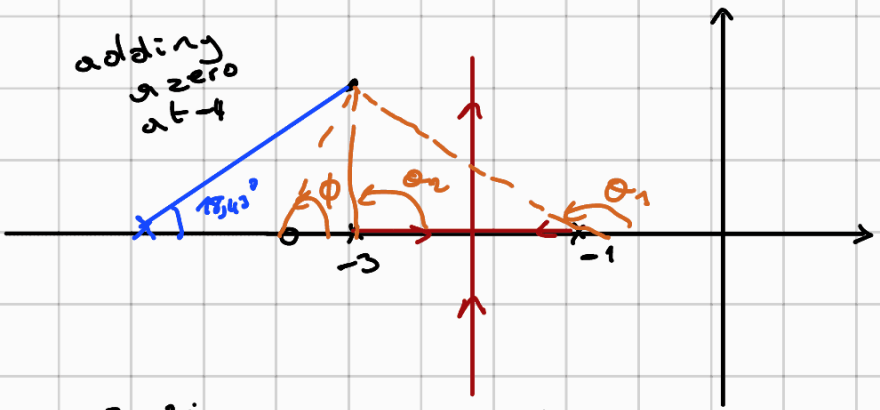
→ lead compensator

Dominant pole = $-3+2j$

for step input ← Steady state error = 10% → lag compensator



$$G_c = \frac{s+\frac{1}{T}}{s+\frac{1}{\alpha T}}$$



$$\angle(-3+2j) = -\theta_1 - \theta_2 + \phi = -(180 - \tan^{-1}(\tfrac{2}{2})) - 90 + \tan^{-1}(\tfrac{2}{1})$$

$$= -135 - 90 + 63,43^\circ = -161,56$$

Phase deficiency $= -135 - 90 + 63,43^\circ = -161,56^\circ$

So compensator $G_c(s) = \frac{s+4}{s+9}$

$$E_{ss} = \frac{1}{1+K_p}$$ → static position error constant

$$K_p = \lim_{s\to 0} G(s)$$

$$E_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)} = \lim_{s\to 0} \frac{1}{1+G(s)} = \frac{1}{1+\lim_{s\to 0} G(s)} = \frac{1}{1+K_p}$$

$G(s) = k_c \left(\frac{s+4}{s+9}\right)\left(\frac{1}{(s+1)(s+3)}\right)$ compensator system

$E_{ss} = 0.1 = \frac{1}{1+k_p} \Rightarrow 1+k_p = 10 \Rightarrow k_p = 9$

$9 = k_p = \lim_{s\to 0} G(s) = k_c \left(\frac{4}{9}\right)\left(\frac{1}{1\times 3}\right)$

$k_c = \left(\frac{4}{27}\right) = 9$

$\Rightarrow k = 9 \times \frac{27}{4}$

$G(s) = \left| k \frac{(s+6)}{(s+9)} \quad \frac{1}{(s+1)(s+3)} \right| = 1$

k give dominant pole